

GUJARAT TECHNOLOGICAL UNIVERSITY**BE - SEMESTER-V (NEW) EXAMINATION – WINTER 2021****Subject Code:3150911****Date:23/12/2021****Subject Name:Power System- II****Time:02:30 PM TO 05:00 PM****Total Marks: 70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

- Q.1**
- (a) Write a short note on Ferranti effect. **03**
- (b) Two single phase loads having impedances $Z_1 = 100 + j80 \Omega$ and $Z_2 = 150 - j60 \Omega$ are connected in parallel and this parallel combination is supplied from a source of 200 V a.c. Calculate: **04**
- (1) Complex power drawn by each load
- (2) Overall power factor of the combination as viewed from the source.
- (c) Explain working of a synchronous generator connected to infinite bus. Draw equivalent circuit and phasor diagram. **07**
- Q.2**
- (a) Explain classification of transmission lines. Define voltage regulation and efficiency of a transmission line. **03**
- (b) A balanced three phase load of 30 MW, 0.85 p.f. lagging is supplied at 132 kV, 50 Hz by overhead transmission line. The series impedance and shunt admittance of the line are $(20 + j52) \Omega$ and $j3.15 \times 10^{-4}$ siemen. Using nominal-T method, determine, **04**
- (i) ABCD constants of the line
- (ii) Sending end voltage
- (c) Derive equations of power flow through a transmission line. **07**
- OR**
- (c) Explain rigorous solution of long transmission line **07**
- Q.3**
- (a) Show that symmetrical component transformation is power invariant. **03**
- (b) Determine symmetrical components of phase-a voltages, **04**
- $V_a = 200 \angle 0^\circ$ V, $V_b = 200 \angle 245^\circ$ V and $V_c = 200 \angle 105^\circ$ V
- (c) Discuss transients on a transmission line in case of a symmetrical fault. **07**
- OR**
- Q.3**
- (a) Draw equivalent circuits of a synchronous machine during (a) sub-transient state (b) transient state and (c) steady state after occurrence of a three phase fault on its terminals. **03**
- (b) Write a short note on selection of circuit breaker. **04**
- (c) A delta connected balanced resistive load is connected across an unbalanced three phase supply as shown in fig. (1). The line currents in phase A and B are given as, $I_A = 10 \angle 30^\circ$ A and $I_B = 15 \angle -60^\circ$ A. Find the symmetrical components of line current I_A and delta current I_{AB} . **07**

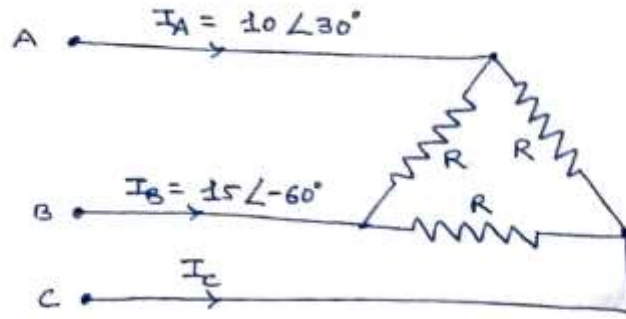


Fig. (1)

- Q.4 (a)** Draw zero sequence network of a three phase transformer with **03**
- (1) Star-Star connection with both neutrals grounded
 - (2) Star delta connection with star side neutral grounded
- (b)** A 3-phase, 11 kV, 10 MVA synchronous generator has sequence reactances of $X_0 = 0.05$ p.u., $X_1 = X_2 = 0.15$ p.u., . Assuming that the generator is initially operating at no load, calculate the magnitude of fault current in case of an L-G fault at generator terminals. **04**
- (c)** Fig.(2) shows a power system network. The system data is given in the table below. **07**

Equipment	MVA Rating	Voltage Rating	X_1 , p.u.	X_2 , p.u.	X_0 , p.u.
Gen-1	100	11 kV	0.25	0.25	0.05
Gen-2	100	11 kV	0.2	0.2	0.05
Transformer-T1	100	11/220 kV	0.06	0.06	0.06
Transformer-T1	100	11/220 kV	0.07	0.07	0.07
Line	100	220 kV	0.1	0.1	0.3

Neutral point of generator-2 is grounded through a reactance of 0.03 p.u. on generator base values.

A single line to ground fault occurs at bus-2 of the network. Find the fault current.

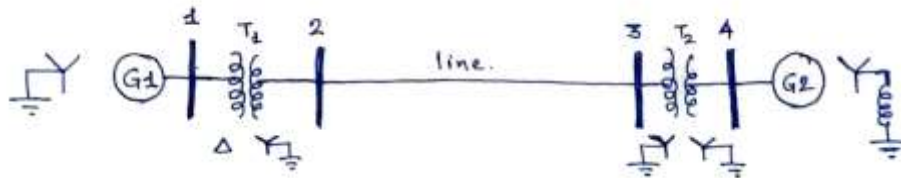


Fig. (2)

OR

- Q.4 (a)** A power system network is shown in the single line diagram of fig. (3). Draw zero sequence network for the system. **03**



Fig. (3)

- (b) Explain analysis of a line to line fault. 04
 (c) A 3-phase synchronous generator rated 10 MVA, 11 kV delivers power to a static load through a line. Positive negative and zero sequence impedances of the generator and line are mentioned in the table below. 07

Equipment	X_1 , p.u.	X_2 , p.u.	X_0 , p.u.
Generator	0.2	0.2	0.05
Line	0.1	0.1	0.3

The load draws 5 MW power at 10 kV at 0.8 p.f. lagging before a line to ground fault takes place at load terminals. Determine the fault current. Assume the generator neutral to be solidly grounded.

- Q.5 (a) What is corona? Write formula of Disruptive critical voltage and briefly describe each variable in the formula. 03
 (b) What is insulation coordination? Define and explain BIL. 04
 (c) What do you understand by “**Travelling Waves**” on transmission lines? Derive equations showing relationship between voltage and current in a forward wave and a backward wave on a lossless transmission line. 07

OR

- Q.5 (a) A 3-phase, 220 kV, 50 Hz transmission line is 250 km long. Calculate the corona loss occurring on the line if the radius of each conductor is 11 mm and Equivalent spacing between conductors is 6 m. It is given that the relative density of air is 0.9456 and disruptive critical voltage is 118 kV per phase. 03
 (b) Discuss recovery transients due to removal of a short circuit. 04
 (c) Explain types of lightning strokes. 07
